

Unit 2. Random Variables and Univariate Distributions

Bayes' theorem, p.m.f., p.d.f. and c.d.f.

Introductory Statistics for Econometrics

Universidad Carlos III de Madrid

Adapted from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 4–6, CC BY-NC-SA 4.0.

Roadmap

- 1 Bayes' theorem
- 2 Random variables
- 3 Discrete random variables and the p.m.f.
- 4 Continuous random variables and the p.d.f.
- 5 The cumulative distribution function
- 6 Wrap-up

From Total Probability to Bayes

Recall. For a partition $\{B_1, \dots, B_n\}$ of $\mathcal{S}$:

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i).$$

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For any partition $\{A_1, \dots, A_n\}$ and event B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i) \mathbb{P}(A_i)}{\sum_j \mathbb{P}(B \mid A_j) \mathbb{P}(A_j)}.$$

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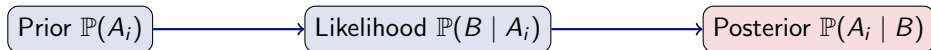
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Why we care. Whenever data B is informative about an unknown state A_i , Bayes turns **prior** + **likelihood** $\longrightarrow$ **posterior**. This is the entire logic of statistical inference.

Vocabulary

- $\mathbb{P}(A_i)$ — the **prior**: belief before seeing the data.
- $\mathbb{P}(B \mid A_i)$ — the **likelihood**: probability of the data under state A_i .
- $\mathbb{P}(A_i \mid B)$ — the **posterior**: updated belief.



Example: a diagnostic test

- Sensitivity: $\mathbb{P}(B \mid A) = 0.99$.
- False-positive rate: $\mathbb{P}(B \mid A^c) = 0.05$.
- Prevalence (prior): $\mathbb{P}(A) = 0.005$.

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Lesson: base-rate neglect

A highly accurate test on a rare disease still produces mostly false positives. People (and pundits) routinely forget this.

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Switch. The host's action is *informative* because it was constrained: he had no choice but to open C if the prize was behind B.

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The probability $\mathbb{P}$ on $\mathcal{S}$ **induces** a distribution for X :

$$\mathbb{P}(X \in A) \equiv \mathbb{P}(\{s \in \mathcal{S} : X(s) \in A\}).$$

In practice we work directly with $\mathbb{P}(X \in A)$ and forget $\mathcal{S}$.

Three families of random variables

- **Discrete:** X takes at most countably many values.
 - ▶ coin flip, dice, number of unemployed in a survey.
- **Continuous:** X takes values in an interval; any single point has probability zero.
 - ▶ asset returns, waiting times, GDP.
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Indicator trick

Any event A becomes a Bernoulli random variable $\mathbf{1}_A \in \{0, 1\}$, $\mathbb{P}(A) = \mathbb{E}[\mathbf{1}_A]$.

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Properties:

- $f_X(x_i) \geq 0$ for every i ;
- $\sum_i f_X(x_i) = 1$;
- $\mathbb{P}(X \in A) = \sum_{x_i \in A} f_X(x_i)$.

Three discrete benchmarks

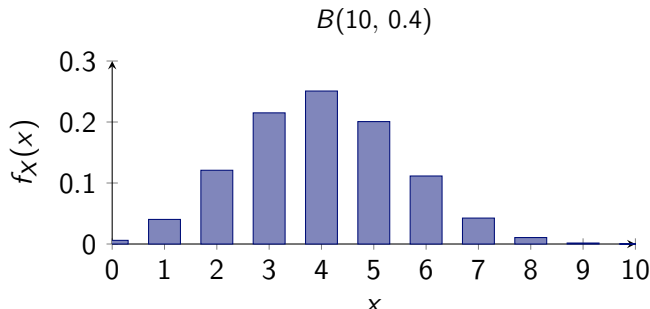
- **Discrete uniform** on $\{x_1, \dots, x_K\}$: $f_X(x_i) = 1/K$.
- **Bernoulli(p)**: $X \in \{0, 1\}$ with $\mathbb{P}(X = 1) = p$. Workhorse for binary outcomes (default, employment, treatment).
- **Binomial $B(n, p)$** : number of successes in n independent Bernoulli(p) trials,

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n.$$

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Example: updating a coin's bias

Two hypotheses:

- A^c : fair coin, $p_F = 0.5$;
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Toss 10 times, observe 8 heads (event B).

$$\mathbb{P}(B \mid A) = \binom{10}{8} (0.8)^8 (0.2)^2, \quad \mathbb{P}(B \mid A^c) = \binom{10}{8} (0.5)^{10}.$$

$$\mathbb{P}(A \mid B) \approx 0.46.$$

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- Posterior probability of bias: 11% $\rightarrow$ 46%.
- Single experiment is informative but not conclusive — repeat to sharpen the posterior.

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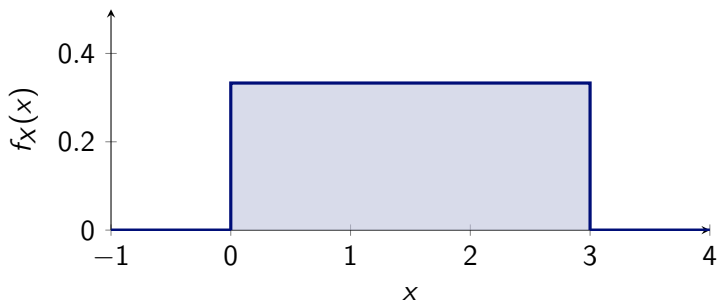
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- $f_X(x) \geq 0$ for all x ;
- $\int_{-\infty}^{\infty} f_X(t) dt = 1$;
- $f_X(x)$ is **not** a probability — it can exceed 1.

Example: continuous uniform $U[0, 3]$



$$f_X(x) = \frac{1}{3} \mathbf{1}_{[0,3]}(x), \quad \mathbb{P}(1 < X < 2) = \int_1^2 \frac{1}{3} dt = \frac{1}{3}.$$

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$$\int_0^3 ax^2 dx = 9a = 1 \implies a = \frac{1}{9}.$$

Step 2: compute a probability.

$$\mathbb{P}(1 < X < 2) = \int_1^2 \frac{t^2}{9} dt = \frac{2^3 - 1^3}{27} = \frac{7}{27} \approx 0.26.$$

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Properties (any c.d.f.)

- 1 $0 \leq F_X(x) \leq 1$.
- 2 F_X is non-decreasing.
- 3 $\lim_{x \rightarrow -\infty} F_X(x) = 0$, $\lim_{x \rightarrow +\infty} F_X(x) = 1$.
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- ④ F_X is right-continuous.

Conversely, any function with these four properties is the c.d.f. of some random variable.

Probabilities from the c.d.f.

$$\mathbb{P}(X > x) = 1 - F_X(x),$$

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a),$$

$$\mathbb{P}(X = x) = F_X(x^+) - F_X(x^-) \quad (\text{jump size}).$$

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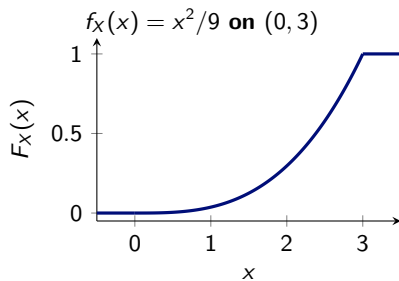
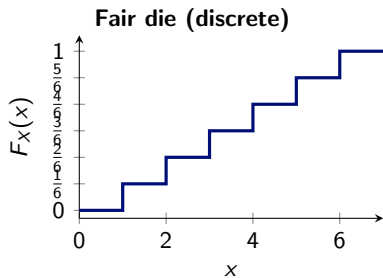
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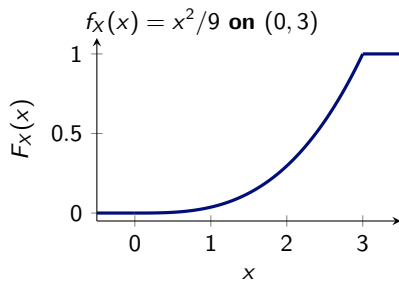
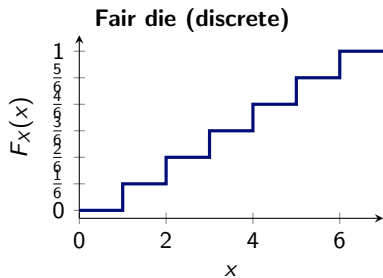
Translation between c.d.f. and p.m.f./p.d.f.

- Discrete: F_X is a **step function**; $f_X(x_i) = F_X(x_i) - F_X(x_i^-)$.
- Continuous: F_X is differentiable a.e. and $f_X(x) = F'_X(x)$.

Example: c.d.f. of a fair die vs. a continuous density



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The discrete c.d.f. has **jumps** of size $1/6$ at each integer; the continuous c.d.f. is smooth.

Take-aways

- ① **Bayes' theorem**: $\text{prior} \times \text{likelihood} \propto \text{posterior}$. Beware base-rate neglect.
- ② A **random variable** is a function $X : \mathcal{S} \rightarrow \mathbb{R}$. We work with its induced distribution $\mathbb{P}(X \in A)$.
- ③ **Discrete RV** $\rightarrow$ p.m.f. $f_X(x) = \mathbb{P}(X = x)$. Bernoulli and Binomial are foundational.
- ④ **Continuous RV** $\rightarrow$ p.d.f. f_X , with $\mathbb{P}(X \in A) = \int_A f_X$.
- ⑤ The **c.d.f.** $F_X(x) = \mathbb{P}(X \leq x)$ unifies both worlds. Jumps = point masses; slopes = density.

Next unit

Joint, marginal and conditional distributions of two or more random variables.

Thank you

Questions?

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